

Barkai Lazimy

Engineering Physics & Math | University of Wisconsin-Madison
PORTFOLIO | <https://bitarmada.github.io/Portfolio/>

439 E Campus Mall
#200
Madison, WI 53703
(608) 733-9598
Blazimy@wisc.edu
Anticipated Graduation Date, May 2028

SKILLS

Programming languages: C, C++, Python, JavaScript, Java, R, HTML/CSS, x86 assembly, WASM, Linux & Windows
Machine learning algorithms & Robotics: Graph Networks (GCN, GAT), Neuroevolution of Augmenting Topologies (NEAT), GANs, CNNs, and custom computer vision models. Experience with PyTorch, TensorFlow, and JAX.
GPU Programming: CUDA, WebGPU compute shaders (WGSL), OpenGL & WebGL graphics programming (GLSL)
Physics Simulation: Hartree-Fock multielectron quantum simulation, Anisotropic Finite Element Method, Topology Optimization
Teamwork: Experience working collaboratively, with strong written and verbal communication skills.

EDUCATION & RELEVANT COURSEWORK

University of Wisconsin-Madison Engineering Physics BS Cumulative GPA: 3.782
Numerical Methods, Machine Learning in Physics, Linear Algebra & Differential Equations, Computational Engineering

EXPERIENCE

Machine Learning & Topological Data Analysis, Bhaskar Lab Madison, WI
Machine Learning Research Assistant June 2026 - Present

- **Neural Architecture:** Developed a Graph Attention, GRU, and Hodgkin-Huxley neural ODE-based model to infer neural connectivity from simulated firing data on Watts–Strogatz networks generated at scale using NEST.
- **Computer Vision:** Worked on a synthetic pipeline for Second Harmonic Generation (SHG) images of cancerous and non-cancerous collagen fibers and image segmentation.
- **High-Performance Computing:** Scaled machine learning training pipelines across CHTC clusters utilizing NVIDIA H200 GPUs and high-throughput distributed computing architectures to handle large datasets.

Computational Bioinformatics, Glukhov Lab Madison, WI
Undergraduate Research Assistant June 2026 - Present

- **RNA sequencing & analysis:** Python and R to process and analyze RNA sequencing data: quantifying gene expression, subpopulation analysis, and dimensionality reduction.
- **Physical Simulation:** Wrote a physics-based tissue deformation simulation using a custom FEM algorithm & JAX to analyze extracellular matrix stiffness.

UW Madison Department of Food Science Madison, WI
Research Intern June 2024 - September 2024

- **Wet Lab Work:** Grew bacterial cultures, sequenced DNA, and ran Polymerase Chain Reactions.
- Discovered and named several previously unknown bacterial strains.

AIAlgo LLC Madison, WI
Website and backend developer September 2022 - December 2022

- **Web and Server Development:** Created a website with a custom UI and 3D animations, along with a secure account system and customer database from scratch to handle payments and subscriptions.

INDEPENDENT PROJECTS & RESEARCH — MORE IN PORTFOLIO

[AI Security System](#) — Robotic Prototype

A prototype for a security product that utilizes non-lethal lasers and audio to deter potential threats. I designed and 3d printed a camera turret capable of locking onto attackers with guns/knives using a microcontroller and a custom YOLOv8 computer vision algorithm. The project won first place in an Entrepreneurship Innovation Plan competition in 2025.

[Path Generation for Autonomous Aircraft in Time-Dependent Flow Fields](#) — Algorithm

An Algorithm that incorporates global hourly wind data to generate the fastest routes between any points on Earth using a field-based LSF approach. The algorithm creates a field of reachable points across the Earth that is advected at every iteration with the time-dependent field and expanded using the aircraft's thrust parameters. The optimal path is traced using the gradient of the reachable time field.

[Topology Optimization for Fluid Flows](#) — AI-based algorithm

An algorithm for optimizing solid topologies in fluid flows, a problem that traditional optimization algorithms struggle to solve. Implemented using a Convolutional Neural Network (CNN) to approximate fluid forces and Python + PyTorch to compute topological gradients.

[Differentiable Anisotropic FEM simulation for 3D Printing Optimization & Graph Convolutional Approach](#) — AI Algorithm

An algorithm for generating strength-optimized 3D-printed layer patterns using a custom differentiable anisotropic finite element method written in Python and JAX. Later trained a Graph Convolutional Network built on the dual graph of the finite element mesh to generate similar optimal patterns using a level set function.